

Week 11 — Feature Engineering and Machine Learning Foundations

Homework, Weekly Quiz, and Complete Solutions

DSP + ML • Senior / First-Year Graduate ECE

Homework — Student Version

Show reasoning, equations, and units where appropriate. Python may verify numerical work, but conceptual explanations must be in your own words.

Problem 1: Mel-scale reasoning [10 points]

Using $m = 2595 \log_{10}(1 + f/700)$: (a) compute the Mel coordinates of 500 Hz, 1000 Hz, and 8000 Hz; (b) compare the Mel differences 500→1000 Hz and 7500→8000 Hz; (c) explain what the comparison says about frequency warping.

Problem 2: Mel filterbank matrix dimensions [10 points]

A 4.0 s clip at 22.05 kHz produces an STFT power matrix with $K=1025$ frequency bins and $T=173$ frames. A Mel bank has $B=64$ filters. State the dimensions of P , H , and $S_{\text{mel}} = H P$. How many scalar values are stored before and after Mel pooling?

Problem 3: Power, log compression, and dB [10 points]

Two Mel-band powers are $P_1=1.0$ and $P_2=0.001$ relative to $P_{\text{ref}}=1$. Compute each in dB using $10 \log_{10}(P/P_{\text{ref}})$. Explain why log compression is useful for visualization and ML.

Problem 4: MFCC interpretation [10 points]

Describe the sequence of operations that produces MFCCs from a waveform. Why is the DCT used after log-Mel energies? Explain why “the DCT perfectly decorrelates every audio dataset” is too strong.

Problem 5: Fixed-size preprocessing [10 points]

An audio clip is stereo, 48 kHz, and 2.7 s long. Describe a deterministic preprocessing pipeline that produces a mono, 22.05-kHz, 4.0-s waveform. How many output samples are there? What information might each step discard?

Problem 6: Feature-shape tradeoffs [12 points]

You compute 20 MFCC coefficients over 173 frames. Compare (a) flattening, (b) time mean only, and (c) time mean + time standard deviation. Give D for each and one advantage/disadvantage.

Problem 7: Audio data leakage [14 points]

A long source recording is segmented into multiple excerpts. A student randomly shuffles excerpts and uses an 80/20 train/test split. Explain how source-level leakage can occur. Propose a safer split and explain why random-split accuracy may be misleading.

Problem 8: Scikit-learn baseline and evaluation [14 points]

Write pseudocode or Python for a RandomForestClassifier baseline. Then describe what a confusion matrix reveals that overall accuracy does not. Name one reason to inspect macro-F1.

Problem 9: Graduate stretch — nested decisions and leakage [10 points]

Choose $n_{\text{mfcc}} \in \{13, 20, 30\}$ and $\text{max_depth} \in \{\text{None}, 10, 20\}$ without using the test fold. Design a train/validation/test procedure, then explain how predefined-fold rotation can give a more reliable estimate.

Homework — Complete Solutions

Solution 1

$m(500) \approx 607.4$ Mel, $m(1000) \approx 1000.0$ Mel, $m(8000) \approx 2840.0$ Mel. $\Delta m(500 \rightarrow 1000) \approx 392.5$ Mel; $\Delta m(7500 \rightarrow 8000) \approx 66.7$ Mel. The same 500-Hz physical separation occupies much less Mel distance at high frequency, so the upper frequency range is compressed relative to low frequency.

Solution 2

P is [1025,173], H is [64,1025], and S_mel is [64,173]. P stores 177,325 values; S_mel stores 11,072 values, about a 16× reduction along the frequency dimension while preserving 173 time frames.

Solution 3

P1 = 0 dB; P2 = -30 dB. Log compression turns multiplicative ratios into additive differences and prevents a few large powers from dominating the numeric/visual range.

Solution 4

Waveform → STFT → power → Mel pooling → log/dB → DCT across Mel bands → retain selected coefficients. The DCT rotates correlated overlapping-band coordinates into cosine-basis directions and often compacts smooth spectral-envelope information into low-order coefficients. Exact decorrelation depends on the data covariance and is not guaranteed.

Solution 5

Downmix stereo to mono, anti-aliased resample 48 kHz→22.05 kHz, then right-pad from 2.7 s to 4.0 s. Output length = $4 \times 22050 = 88,200$ samples. Downmixing discards spatial/channel differences; resampling discards content above the new Nyquist limit; padding adds silence. Truncation, for longer clips, would discard late-time content.

Solution 6

Flatten: $D=20 \times 173=3460$; preserves frame order but is high dimensional/alignment sensitive. Mean: $D=20$; compact and time-shift tolerant but destroys order and variability. Mean+std: $D=40$; preserves average and variation but still discards sequence order.

Solution 7

Segments from one source share microphone response, background, source identity, and possibly overlapping acoustic content. If one source appears in train and test, the model can exploit source-specific cues. Group all excerpts from a source into one split; for UrbanSound8K, honor predefined folds. Random-split test accuracy can therefore be optimistic.

Solution 8

Example: `clf=RandomForestClassifier(n_estimators=300,random_state=42,n_jobs=-1); clf.fit(X_train,y_train); y_pred=clf.predict(X_test)`. Accuracy gives total correctness; a confusion matrix shows which true classes are confused with which predictions. Macro-F1 gives each class equal weight and is useful when classes are imbalanced or minority-class performance matters.

Solution 9

Use training to fit each candidate and validation to select `n_mfcc/max_depth`. After choosing, optionally refit on train+validation and evaluate once on untouched test. For stronger estimates, rotate outer held-out predefined folds and repeat model selection entirely inside the remaining folds, then aggregate outer metrics.

Weekly Quiz — Student Version

Suggested time: 15 minutes. Total: 20 points.

Problem 1: Mel feature concept [4 points]

In one or two sentences, explain what the Mel filterbank does to an STFT power spectrum and what axis it preserves unchanged.

Problem 2: MFCC reasoning [4 points]

Put in order: DCT, STFT power, log/dB, Mel filterbank. State one reason MFCCs may be attractive for a classical classifier.

Problem 3: Shape calculation [3 points]

A 20×173 MFCC matrix is summarized by mean and standard deviation of each coefficient across time. What is D?

Problem 4: Leakage [5 points]

Why is randomly splitting segmented audio excerpts potentially invalid even if train and test contain no identical files?

Problem 5: Scikit-learn API [4 points]

Fill in: instantiate a classifier, _____ it on $(X_{\text{train}}, y_{\text{train}})$, then _____ labels for X_{test} . Name one evaluation tool besides overall accuracy.

Weekly Quiz — Solutions and Grading Guide

Solution 1

It computes overlapping weighted sums of neighboring STFT power bins on a perceptual-frequency scale. It changes/compresses the frequency axis while preserving the time-frame axis.

Solution 2

STFT power $\rightarrow$ Mel filterbank $\rightarrow$ log/dB $\rightarrow$ DCT. MFCCs give compact coefficients that often summarize spectral-envelope information and work naturally as classical tabular features.

Solution 3

$D = 20$ means $+ 20$ standard deviations $= 40$.

Solution 4

Different excerpts can come from the same physical recording and share source identity, sensor response, background, or overlapping content. The test set is then not source-independent, making generalization appear better than it is.

Solution 5

fit; predict. Examples: confusion matrix, precision, recall, F1, or classification report.

Instructor Notes

Reward explanations that identify what each preprocessing step preserves and discards. For graduate students, require explicit leakage reasoning and a distinction between training-fitted transformations, validation-driven choices, and final test evaluation.